Unifying FEGA Metadata: A Standardized Model for Multimodal Data

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COMMUNITY AND VISION

The **Federated EGA** (FEGA) already links a growing network of national human-data nodes under one discovery umbrella, giving researchers policy-compliant access to **sensitive omics data** without moving the files across national borders.

Our **Metadata Working Group** brings together 16 experts from seven European institutes to build a **common metadata model** and **validation toolchain**, keeping FEGA data FAIR and research-ready. European projects such as GDI feed in real-time requirements.

We would like to extend an **open invitation** to all GDI and ELIXIR nodes, GA4GH driver projects and beyond to engage in our discussions and shape the FEGA model.

Federated European Genome-phenome Archive

GROWING SCOPE

We began with **EGA's core genomic datasets** and are now progressively broadening our reach to cover additional data domains within the Federated Human Data (FHD) ecosystem:

1. Genomics
2. Microarrays
3. Proteomics
4. Microbiome
5. Phenotypes
6. Imaging
7. Tra
8.

Additional data modalities will follow in **phased releases**, prioritised according to real-world demand and the readiness of community contributors.

Core Model

Organisational context that provides the **provenance** and **grouping**

Administrative

Needed for tracking and reporting

Information about **any biological material**

Biomaterial

From whole organisms to subcellular components and macromolecules

Used **protocols**

Process

And their **execution** details.

Resource containing **data**

Data File

FASTQ, BAM, DICOM, GFF, VCF, Phenopacket, TSV, BCF, MGF, FASTA, CEL, BED, gVCF, mzTab

Governance layer defining **data access control**

Data Management

How data is processed, protected and shared

A GLIMPSE OF SERIALISATION

Validation rules expressed in JSON Schema, enriched with a `@context` so any record expands to RDF

JSON-LD

JSON Schema

```
1  "birthPlace": {
2    "@context": "http://schema.org/birthPlace",
3    "type": "object",
4    "meta:sssomMappings": [{
5      "object_id": "HMMD.dataset.v1.Subject.'country of origin'"
6    }],
7    "properties": {
8      "id": {
9        "graphRestriction": {
10         "ontologies": ["obo:gaz"],
11         "classes": ["NCIT:C25464"],
12         "relations": ["rdfs:subClassOf"]
13       }
14     }
15   },
16   "allOf": [{
17     "$ref":
18 "https://raw.githubusercontent.com/ga4gh-beacon/beacon-v2/main/
     .../geographicOrigin"
19   }]
20 }
```

Biovalidator queries live APIs during validation, allowing dynamic and easy-to-maintain constraints

Biovalidator

ENA OLS

All schemas, tests and docs live in an open GitHub repository, ensuring full transparency and reuse

Embedded SSSOM tables map every field across community vocabularies (e.g., GDI or Phenopackets)

European Genomic Data Infrastructure

ELIXIR HDTR WP4 & WP8

`$ref` links pull terms straight from other standards (e.g., Beacon-v2), ensuring instant alignment

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For further details, please follow these QR codes to the respective resources

FEGA Metadata GitHub | FEGA Onboarding | Public Presentation | Group Slack Channel

ELIXIR ALL HANDS 2025

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